Quantitative Financial Risk Management

Duisenberg Honours Programme in Quantitative Risk Management

Period 4.5

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Part III

Market Risk Management

(McNeil, Frey, and Embrechts, 2015, Ch. 2B)

1 Motivation

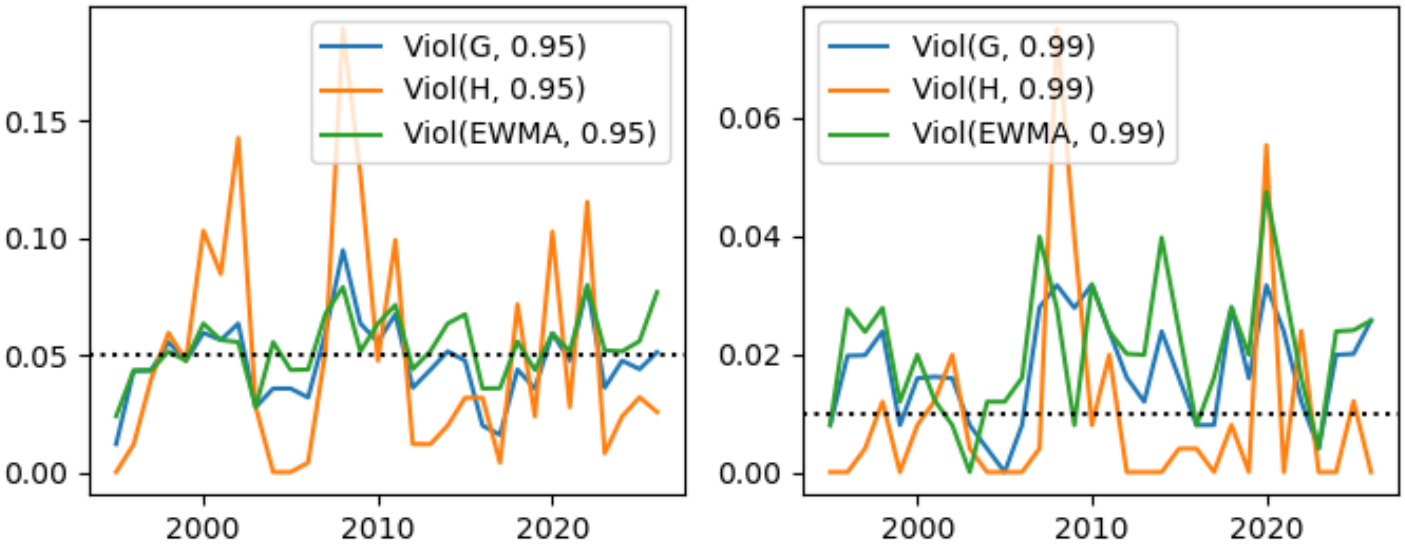
2 Backtesting VaR and ES

3 Stress testing

4 Market Risk

5 Review

Are we measuring risk correctly? How can we manage risk?



Does VaR and/or ES make sense?
Check on historic data!

1. In-sample: Estimation over $[1, \ldots, T]$, check over $[1, \ldots, T]$
2. Out-of-sample:
   1. Estimation over $[1, \ldots, T_1]$, check over $[T_1 + 1, \ldots, T]$ (fixed window)
   2. Set $j = 0$, estimate over $[j + 1, j + T_0]$, check over $[j + T_0 + 1, \ldots, j + T_0 + h]$, increase j and repeat until $j + T_0 + h = T$ (moving/rolling window).

What is better?

- Length of sample sufficient for estimation
- Length of sample sufficient for backtesting
- Computational load acceptable

$\Rightarrow$ Depends...

How to test VaR model? What do we want from a good VaR model?

1. that losses exceed VaR(α) approximately $1 - \alpha$ of the time
2. that losses exceed VaR in a "random" fashion, and not in big clusters

Ad-hoc testing:

1. Count fraction of violations $r_t > \mathrm{VaR}_t(\alpha)$ in historical dataset, compare with $1 - \alpha$
2. Graph the VaR violations: Do they occur 'randomly', or are 'clusters of violations' visible?

Model	Level	Violations
GARCH	0.95	0.047194
Historic	0.95	0.050000
EWMA/RM	0.95	0.052934
GARCH	0.99	0.017092
Historic	0.99	0.010077
EWMA/RM	0.99	0.020408

Table: 1995-2026 GARCH/Historic/RM($\lambda = 0.94$) comparison (single estimation, in-sample comparison)

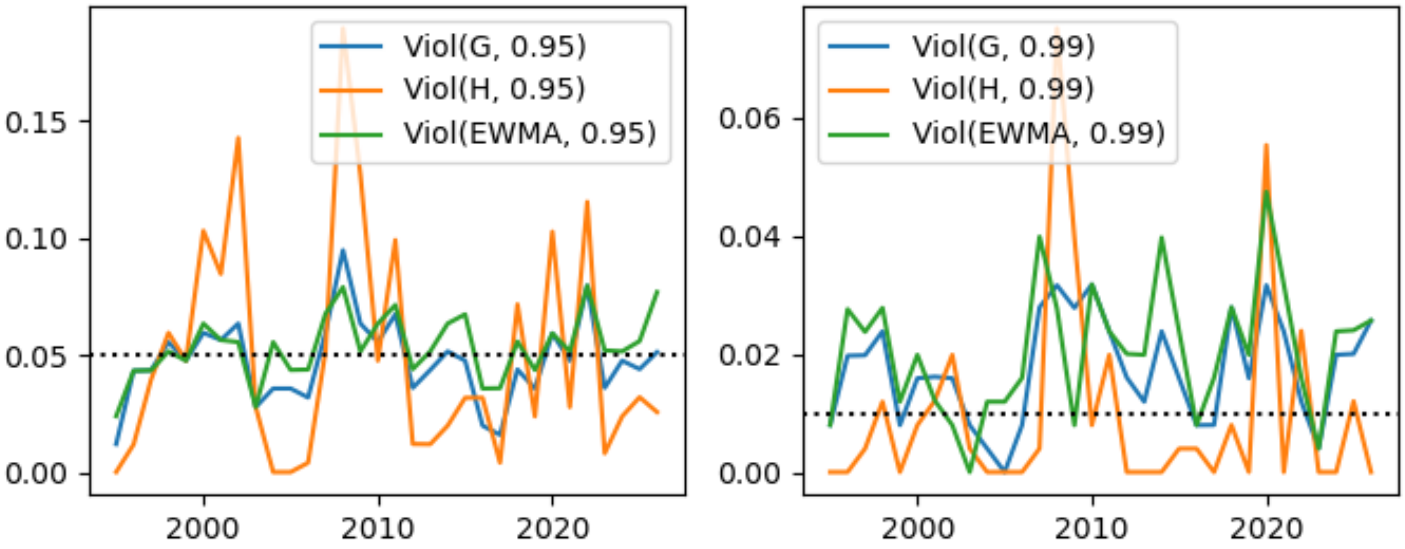


Figure: 1995-2026 GARCH/Historic/RM($\lambda = 0.94$) comparison (single estimation, in-sample comparison, yearly tests)

Statistical testing of VaR:
Define VaR violation indicator,

$$I_t = 1_{\{L_{t+1} > \mathrm{VaR}_\alpha^t\}}$$

If the VaR model is specified correctly:

$$H_0 : I_t \equiv 1_{\{Z > q_\alpha(F_Z)\}} \sim \text{i. i. d. Bern}(1-\alpha)$$

In practice: Use violation indicators,

$$\hat{I}_t = 1_{\left\{L_{t+1} > \widehat{\mathrm{VaR}}_\alpha^t\right\}}$$

and check against Bern$(1-\alpha)$, or the $I_c = \sum \hat{I}_t \sim \mathrm{Bin}(T, 1-\alpha)$

Test setup (5-step plan Business Statistics):

1. Use hypothesis $H_0 : I_c = I_0$ vs $H_1 : I_c \neq I_0$. Test at a level of e.g. $\alpha_c = 0.05$.
2. The test statistic here is $\hat{I}_c = \sum_{t=1}^{T} \hat{I}_t$, the total number of violations, and $I_0 = T(1 - \alpha_{\mathrm{VaR}})$ is the expected number of violations, at VaR of level α_{VaR}.
   The rejection region is for large and small values of $\hat{I}_c$.
3. Under $H_0 : I_c \sim \mathrm{Bin}(T, 1 - \alpha_{\mathrm{VaR}})$ (no further conditions needed)
4. Calculate $\hat{I}_c$ from the data, and find $c_L = q_{\alpha_c/2}(F_I), c_U = q_{1-\alpha_c/2}(F_I)$, the 2.5% and 97.5% critical values of the binomial distribution
5. Draw your conclusion: If $\hat{I}_c \notin [c_L, c_U]$, reject the H_0

Critical values of the binomial distribution are hard to find sometimes...
Hence:

- If $T\alpha_{\mathrm{VaR}} \geq 10$ and $T(1-\alpha_{\mathrm{VaR}}) \geq 10$ (expected number of violations/non-violations at least 10)

then

$$I_c \sim \mathrm{Bin}(T, 1-\alpha_{\mathrm{VaR}}) \overset{\text{approx}}{\sim} \mathcal{N}(T(1-\alpha_{\mathrm{VaR}}), T\alpha_{\mathrm{VaR}}(1-\alpha_{\mathrm{VaR}}))$$

so use critical values c_L and c_U from the normal density instead.
Alternatively, in Step 4, calculate the p-value using the normal density, and reject if p-value $< \alpha_c$.

Suppose you have 4 years of daily observations, $T = 1000$, and test a $\alpha_{\mathrm{VaR}} = 99\%$, with $\hat{I} = 15$ daily violations in the sample.
Q: Is this a significant result, at a common testing level of α_c?

Suppose you have 4 years of daily observations, $T = 1000$, and test a $\alpha_{\mathrm{VaR}} = 99\%$, with $\hat{I} = 15$ daily violations in the sample.
Q: Is this a significant result, at a common testing level of α_c?
A: See test schedule above. Using $\alpha_c = 0.05$, we find $c_L = 4 = q_{2.5\%}(F_I), c_U = 17$. So a value of $\hat{I}_c = 15$ is not rejected at the 5% level.
The two-sided p-val according to the binomial test is $p = 0.096$
Alternatively, as $T\alpha_{\mathrm{VaR}} = 10 \geq 10$, the normal approximation may (just!) be used. The $\hat{Z} = \frac{\hat{I}_c - T(1-\alpha_{\mathrm{VaR}})}{T\alpha_{\mathrm{VaR}}(1-\alpha_{\mathrm{VaR}})} = 1.59$, leading to a two-sided p-val of $p = 0.11$.

Suppose you have 4 years of daily observations, $T = 1000$, and test a $\alpha_{\mathrm{VaR}} = 99\%$, with $\hat{I} = 15$ daily violations in the sample.
Q: Is this a significant result, at a common testing level of α_c?
Q2: Why is this 0.096 (or 0.11!) the p-value? What is the conclusion?

If the VaR violations are indeed independent, then

- the time between VaR events will be geometrically distributed (McNeil, Frey, and Embrechts, 2015, §9.3.1)
- and this geometric distribution, as the events are rare, can then be approximated by a (continuous) exponential distribution, which is easier to work with.

Assume I violations occur at times $t \in \{T_1, T_2, \ldots, T_I\}$, with $T_0 \equiv 0$.
Define spacings $S_j = T_j - T_{j-1}$. Then

$$S_j \sim \text{i.i.d.}\,\mathrm{Geom}(p = 1 - \alpha_{\mathrm{VaR}}), \quad E(S) = \frac{1}{p}, \quad \mathrm{Var}(S) = \frac{1-p}{p^2}$$

$$\Leftrightarrow S_j \overset{\text{approx}}{\sim} \mathrm{Exp}(\lambda = p) \qquad E(S) = \frac{1}{\lambda}, \quad \mathrm{Var}(S) = \frac{1}{\lambda^2}$$

To test:

1. Make a QQ plot of spacings S against Geometric or Exponential distribution
2. Do a LR test of Exponential spacings against the Weibull alternative (advanced...)

Table: Backtesting VaR counts

Model	Level	Count	cL	cU	p
GARCH	0.95	370	355	430	0.2644
Historic	0.95	392	355	430	0.9731
EWMA/RM	0.95	415	355	430	0.2247
GARCH	0.99	134	62	96	0.0000
Historic	0.99	79	62	96	0.8862
EWMA/RM	0.99	160	62	96	0.0000

Q: For the historic model, there is no rejection. Is that logical?

How did that backtesting work?

Example: With GARCH, there were 357 losses larger than VaR(.95)

```
dAlpha= .95
dAlphaTest= .05
vS= np.sqrt(vS2)          # Get volatility
vVaR= VaR(dAlpha, r= dMu, s= vS)     # Get VaR from normal model, with GARCH vol
vI= vLoss > dVaR          # Find violations
iCount= vI.sum()          # Count violations
iN= len(vI)               # Check number of obs
dPl= st.binom.cdf(iCount, iN, 1-dAlpha)    # Check P(T < t|n, p)
dP= 2*min(dPl, 1-dPl)     # Rebuild to p-value
print (f'p-value: {dP}, rej= {dP < dAlphaTest}')
```

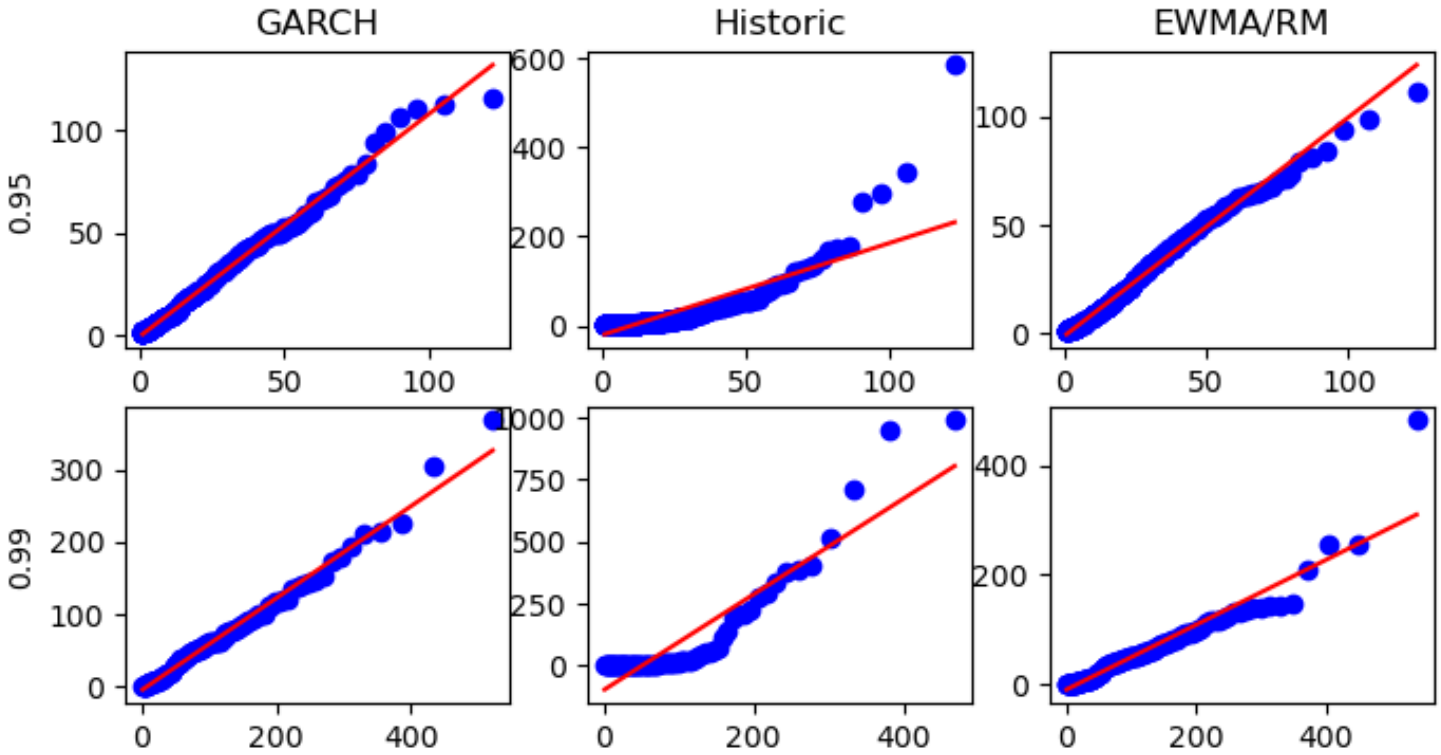


Figure: Spacings of violations vs Geometric$(1 - \alpha_{\mathrm{VaR}})$ distribution

Backtesting VaR Spacing: How?

How did that backtesting/spacing work? Bit of a hassle...

```
dAlpha= 0.95
vViol= dfViol[f'Viol(G, {dAlpha})'].copy()   # Get violations
iN= len(vViol)
vViol.index= np.arange(iN)                   # Set index 0, 1, 2, ...
vC= [0] + list(vViol[vViol].index)           # Get 0 + indices of violations
vSp= np.diff(vC)                             # Get spacing

# Get probability plot
plt.figure(figsize= (8, 4))
st.probplot(vSp, sparams= (1-dAlpha,), plot= plt, dist= 'geom')
plt.show()
```

How to test ES? Cf. VaR:

- Compute loss on the days that VaR is exceeded and compare it to the estimated ES
- Create a rescaled violation residual, to adapt for scale: $K_{t+1} = (L_{t+1} - ES^t)/(ES^t - \overline{L}_t) I_{L_{t+1} > \mathrm{VaR}^t}$
- Significance in difference between the loss and ES can be established by a regular t-test on the violation residual, because under null hypothesis it has mean zero.

MFE §9.3.2

Test setup (5-step plan Business Statistics) for mean shortfall:

1. Use hypothesis $H_0 : \mathrm{ES} = \mathrm{ES}_0$ vs $H_1 : \mathrm{ES} \neq \mathrm{ES}_0$. Test at a level of e.g. $\alpha_c = 0.05$.
2. The test statistic here is

$$t = (\overline{K} - 0)/s(\overline{K})$$

   with ingredients defined on the next page.
   The rejection region is for large and small values of t.
3. Under $H_0 : t \sim t(\nu = I_v)$, with I_v the number of violations
4. Calculate $\hat{t}$ from the data, and find critical values and/or the p-value from the t distribution.
5. Draw your conclusion: If $\hat{t} \notin [c_L, c_U]$, reject the H_0

Elements in the ES test:

- $I_t = I_{L_{t+1} > \mathrm{VaR}_\alpha^t}$ is the violations indicator,
- $I_v = \sum I_t$ the number of shortfalls,
- $K_{t+1} = \left(\frac{L_{t+1} - \mathrm{ES}_\alpha^t}{\mathrm{ES}_\alpha^t - \mu_{t+1}} \right) \times I_t$ the rescaled violation residual (which is zero when there is no violation)
- with $\overline{K}_{t+1} = \sum K_{t+1} / I_v$ the average shortfall, averaged only over the violation days,
- and ES_0 is the expected shortfall that the model predicted over this period.
- The $s^2(\overline{K}) = \frac{1}{I_v} s^2(K)$ is the variance of the observed *average* violation residual,
- where $s^2(K)$ is the variance of the shortfalls themselves again *excluding the non-violation shortfalls*.

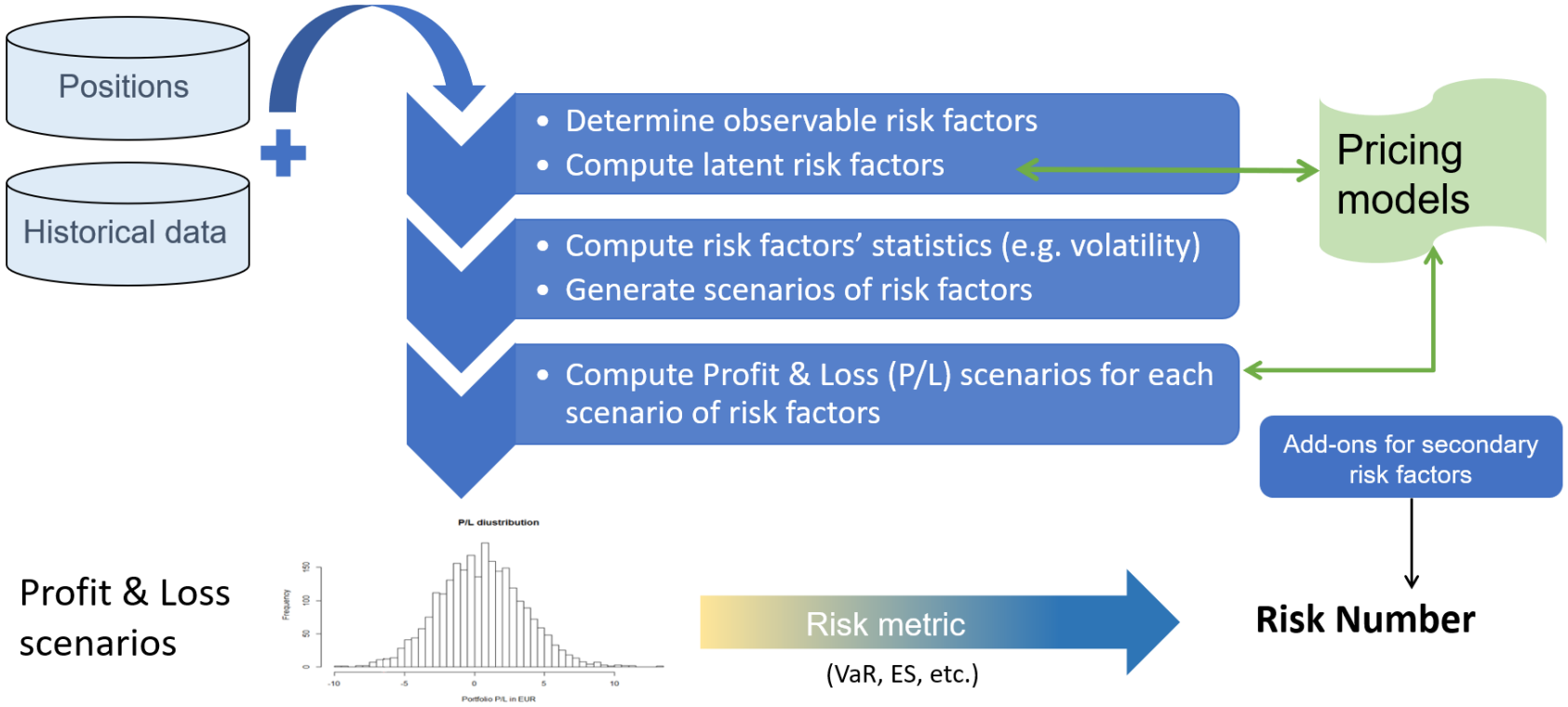
Q: Is the average residual $\overline{K}$ correctly calculated here?

Nowadays VaR and ES must be augmented with stress scenarios.
European Banking Association (EBA): Stressed VaR
Long story short:

- Some freedom for banks
- ... but method should be well-defined, defendable
- Have to find 'stress period', when movements were largest
- Generally, 12 months to be chosen, by risk factors or historic VaR computations
- Or find 'stress scenarios', daily loss of all equity markets of 10% (or more), or daily shock to exchange rate of 20%.
- Results incorporated into current VaR estimation, simple example:

$$\mathrm{VaR}(\text{total}) = 0.75\mathrm{VaR}_t + 0.25\mathrm{VaR}_{\text{stress}}$$

MR: Market Risk Model Setup



To discuss!

A portfolio on market risk may consist of one or more elements of:

- Stock + Stock Indices
- Bonds, loans, fixed income products
- Foreign currency/foreign exchange risk
- Commodities and commodity indices
- Other risky assets: Real Estate Indices, Private Equity Portfolios/Funds, other ETFs

(repeat) Definitions:
Consider

- portfolio with value V_t,
- observable at time t
- looking at risk over $[t, t+1]$
- where $V_{t+1|t}$ is unknown/random.
- $L_{t+1} = -(V_{t+1} - V_t)$ is the loss, with loss distribution,
- and we call $-L_{t+1}$ the profit, or profit-and-loss, with P&L distribution.

(note we look at *loss= direct difference* here, not log-differences or percentage differences)

Portfolio value V_t is not 'just a random variable', but depends on

a. time t (or $\tau_t \equiv t(\Delta t)$)
b. vector $\boldsymbol{Z}_t$ of risk factors $\boldsymbol{Z}_t = (Z_{t1}, \ldots, Z_{td})'$
c. mapping $f(\cdot)$ (or $g(\tau_t, \boldsymbol{Z}_t)$)

such that

$$V_t = f(t, \boldsymbol{Z}_t) \tag{3.1}$$

Risk factors could be

1. prices of financial assets
2. yields
3. exchange rates

(Use τ_t to explicitly move to discrete time increments of size Δt)

Define risk factor changes as

$$\boldsymbol{X}_t = \boldsymbol{Z}_t - \boldsymbol{Z}_{t-1}$$

Generally, you'd expect a link between changes in risk factors and changes in portfolio value, hence

$$\begin{aligned} L_{t+1} &= -(V_{t+1} - V_t) = -(f(t+1, \boldsymbol{Z}_{t+1}) - f(t, \boldsymbol{Z}_t)) \\ &= -(f(t+1, \boldsymbol{Z}_t + \boldsymbol{X}_{t+1}) - f(t, \boldsymbol{Z}_t)) \end{aligned} \tag{3.2}$$

Q: What is known in $\mathcal{F}_t$? What is known only in $\mathcal{F}_{t+1}$?

As indeed $\boldsymbol{Z}_t|\mathcal{F}_t$ is known, say $\boldsymbol{z}_t$, we can rewrite the loss function

$$L_{t+1} = -\left(f(t+1, \boldsymbol{Z}_t + \boldsymbol{X}_{t+1}) - f(t, \boldsymbol{Z}_t)\right) \tag{3.2}$$

into the loss operator

$$l_{[t]}(\boldsymbol{x}) = -(f(t+1, \boldsymbol{z}_t + \boldsymbol{x}) - f(t, \boldsymbol{z}_t)), \qquad \boldsymbol{x} \in \Re^d \tag{3.3}$$
$$\Leftrightarrow L_{t+1} \equiv l_{[t]}(\boldsymbol{X}_{t+1})$$

This means:

- Loss l over time period $[t, t+1)$ is a function of the current, known risk factors $\boldsymbol{z}_t$, and a random change in (multivariate) risk factors $\boldsymbol{x}$
- If the random change $\boldsymbol{x}$ comes out as $\boldsymbol{X}_{t+1}$, then the earlier loss notation L_{t+1} results

MFE §9.1.1

Example: Loss operator Portfolio

Consider a portfolio of d stocks, with α_i shares per stock (fixed), and stock price $S_{t,i}$, for stock $i = 1, \ldots, d$.
Then we get

- Risk factors: $Z_{t,i} = \log S_{t,i}$, logarithmic prices
- Risk factor changes: $X_{t+1,i} = \log S_{t+1,i} - \log S_{t,i}$, the log-returns
- Mapping: $V_t = f(t, \boldsymbol{Z}_t) = \sum_{i=1}^{d} \alpha_i S_{t,i} = \sum_{i=1}^{d} \alpha_i e^{Z_{t,i}}$
- ...

Example: Loss operator Portfolio

- ...
- Loss:

$$L_{t+1} = -\left(\sum_{i=1}^{d} \alpha_i e^{Z_{t+1,i}} - \sum_{i=1}^{d} \alpha_i e^{Z_{t,i}}\right)$$
$$= -V_t \sum_{i=1}^{d} \omega_{t,i} \left(e^{X_{t+1,i}} - 1\right)$$

with $\omega_{t,i} = \alpha_i S_{t,i} / V_t$

- Loss operator:

$$l_{[t]}(\boldsymbol{x}) = -V_t \sum_{i=1}^{d} \omega_{t,i} \left(e^{x_i} - 1\right)$$

Note that the loss operator

$$l_{[t]}(\boldsymbol{x}) = -(f(t+1, \boldsymbol{z}_t + \boldsymbol{x}) - f(t, \boldsymbol{z}_t)), \tag{3.3}$$

may be very non-linear, through its mapping function f.
Use Taylor expansion on mapping function $f(t+1, \boldsymbol{z}_t + \boldsymbol{x})$:

$$\begin{aligned} f(t+1, \boldsymbol{z}_t + \boldsymbol{x}) &\approx f(t, \boldsymbol{z}_t) + f_t(t, \boldsymbol{z}_t) \times \Delta t + \sum f_{z_i}(t, \boldsymbol{z}_t) x_i \\ \Leftrightarrow l_{[t]}(\boldsymbol{x}) &= -(f(t+1, \boldsymbol{z}_t + \boldsymbol{x}) - f(t, \boldsymbol{z}_t)) \\ &\approx -\left(f_t(t, \boldsymbol{z}_t)\Delta t + \sum f_{z_i}(t, \boldsymbol{z}_t) x_i\right) = l^{\Delta}_{[t]}(\boldsymbol{x}) \end{aligned}$$

with f_t the partial derivative w.r.t. t, and f_{z_i} likewise w.r.t. z_i.

Hence linearised loss operator $l^{\Delta}_{[t]}(\boldsymbol{x})$ is

$$l_{[t]}(\boldsymbol{x}) \approx l^{\Delta}_{[t]}(\boldsymbol{x}) = -\left(f_t(t, \boldsymbol{z}_t)\Delta t + \sum f_{z_i}(t, \boldsymbol{z}_t)x_i\right)$$

Note:

- $f_t(t, \boldsymbol{z}_t)$ may often be small (small time steps) or even zero (no explicit dependence on time)
- linearisation convenient in analysis
- works best if loss operator is nearly linear/time steps small
- crucial for variance-covariance method
- could be extended to the quadratic loss operator, using the second order Taylor expansion

Example: Linearised Loss operator Portfolio

Take the portfolio mapping:

$$V_t = f(t, \boldsymbol{Z}_t) = \sum_{i=1}^{d} \alpha_i e^{z_{t,i}}$$

First derivatives of mapping:

- $f_t(t, \boldsymbol{z}_t) = 0$
- $f_{z_i}(t, \boldsymbol{z}_t) = \alpha_i e^{z_{t,i}}$, $i = 1, \ldots, d$

giving linearised loss

$$L^{\Delta}_{t+1} = -\sum_{i=1}^{d} \alpha_i e^{z_{t,i}} X_{t+1,i} = -V_t \sum_{i=1}^{d} \omega_{t,i} X_{t+1,i}$$

Or: Loss is approximately the sum of weighted returns

Example: European Call Option

Suppose the portfolio consists of a single standard European call on stock S (no dividends), with maturity T, exercise price K. According to Black-Scholes,

$$V_t = C^{BS}(t, S; r, \sigma) = S\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2),$$

$$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \qquad d_2 = d_1 - \sigma\sqrt{T-t},$$

with $\Phi(\cdot)$ the standard normal cdf, r the risk-free interest rate, σ the volatility of the stock.

Q: What are risk factors here? Is r truly constant? Is σ?

Example: European Call Option II

In summary:

- Risk factors: $\boldsymbol{Z}_t = (\log S_t, r_t, \sigma_t)'$
- Risk factor changes:
  $\boldsymbol{X}_t = (\log S_t - \log S_{t-1}, r_t - r_{t-1}, \sigma_t - \sigma_{t-1})'$
- Mapping: $V_t = f(t, \boldsymbol{Z}_t) = C^{BS}(t\Delta, S_t; r_t, \sigma_t)$
- Linearised Loss:
  $L^{\Delta}_{t+1} = -\left(f_t(t, \boldsymbol{Z}_t)\Delta + \sum_{i=1}^{3} f_{z_i}(t, \boldsymbol{Z}_t)X_{t+1,i}\right)$

or, more explicit:

$$L^{\Delta}_{t+1} = -\left(C^{BS}_t \times \Delta + C^{BS}_S S_t X_{t+1,S} + C^{BS}_r X_{t+1,r} + C^{BS}_\sigma X_{t+1,\sigma}\right)$$

Here The Greeks are seen, where

- C^{BS}_t is known as the theta of the option
- C^{BS}_S is the delta
- C^{BS}_r is the rho
- C^{BS}_σ is the vega

For later, we would have to look at

- Stocks denominated in foreign currency
- Bond portfolios, long and short bond positions (money in the bank or a loan-leveraged portfolio)
- Combinations

Study the following chapters/sections of the book:

- Remainder of Ch. 2
- §9.1, risk factors
- §9.2, market risk measurement
- §9.3, backtesting, including §9.3.4 for assignment

McNeil, Alexander J., Rüdiger Frey, and Paul Embrechts (2015). *Quantitative Risk Management: Concepts, Techniques, and Tools*. Revised. Princeton: Princeton University Press. ISBN: 978-0-691-16627-8.